

Coding & Solution

Date	30 June 2026
Team ID	xxxxxx
Project Name	Rising Waters
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/mamidalaswathi2006-alt/Rising_Waters_Project
Programming Language(s)	Python, JavaScript, HTML5, CSS3
Framework(s) Used	FastAPI (Python), React.js, Tailwind CSS, Node.js (Express)

Key Features Implemented	1. Real-time telemetry ingestion from regional IoT river-gaugesensors. 2. Machine Learning predictive engine running precipitation andstorm risk evaluation models. 3. Interactive web dashboard rendering dynamic risk heatmapsand historical charts. 4. Automated emergency siren triggers and instant Twilio mobileSMS notification alerts.
Pending / Incomplete Features	1. Advanced multi-language voice-call broadcast system for local regional community alerts. 2. Dynamic offline maps sync features for rescue responders during live network failures.
Setup / Run Instructions	1. Clone repository: git clone <repo_url> 2. Backend setup: Navigate to /backend, run 'pip install -rrequirements.txt' and 'fastapi dev main.py' 3. Frontend setup: Navigate to /frontend, run 'npm install' and'npm run dev' 4. Access dashboard at http://localhost:5173

Code Quality Checklist

S.N o	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes

3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The "Rising Waters" system has been completely configured and locally tested with standard simulated IoT telemetry benchmarks. The Python ML predictive models show 93% accuracy in estimating flood windows. Code layout rigidly adheres to production standards, and all core components are modularly optimized for smooth deployment transitions.